YOUSSEF ISMAILI

ENSEEIHT ENGINEERING STUDENT — 3EA MAJOR

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First-year engineering student at ENSEEIHT in the **Electronics, Electrical Energy and Automatic Control (3EA)** department, following two years of intensive scientific preparatory classes (MP) at LYDEX. Rigorous and passionate about physical modeling, control systems, and systems programming (C, Python, MATLAB). Actively seeking stimulating engineering projects and technical internship opportunities.

EDUCATION

Master's Degree in Engineering (3EA)
ENSEEIHT — Toulouse
2026 — 2029 (Expected)
Major in Electronics, Electrical Energy & Automatic Control.

CPGE MP (Mathematics - Physics)
LYDEX — Benguerir, Morocco
2024 — 2026
Intensive scientific preparatory classes for elite engineering schools. Advanced training in mathematics, physics, and engineering sciences.

SKILLS

Programming Languages:
C | Python | MATLAB

Engineering & Sciences:
Control Systems | Electronics | Signal Processing | Thermodynamics

Tools & Environments:
Simulink | Git / GitHub | GCC | Linux

LANGUAGES

Arabic	**Native**
French	**Fluent (C2)**
English	**Advanced (B2/C1)**

INTERESTS

Football	Team spirit
Volleyball	Coordination
Swimming	Endurance & Discipline

PROJECTS & ENGINEERING EXPERIENCE

Thermodynamic Modeling of the Atmosphere
Scientific Research Project (TIPE) — CPGE LYDEX

- Investigated atmospheric circulation modeled as a heat engine to evaluate the thermodynamic limits of wind energy.
- Designed and operated a convective chimney experimental setup to validate heat transfer laws.
- Conducted a comparative analysis of the Carnot and Curzon-Ahlborn models and simulated thermodynamic cycles using Python.

Technologies & Keywords: Python (NumPy, Matplotlib), Thermodynamics, Scientific Modeling, Energy.

Game Engine & Modular Architecture in C
Academic Project — ENSEEIHT (1A 3EA)

- Designed a complete Snake game engine using structured C programming.
- Implemented dynamic data structures, rigorous pointer management, and memory allocation.
- Developed real-time collision detection algorithms and game state management.

Technologies & Keywords: C Language, Algorithms, Data Structures, GCC, Makefile.

Systems Modeling & Analysis (3EA)
Practical Work & Engineering Projects — ENSEEIHT

- Performed spectral analysis, stability studies, and modeling of continuous linear feedback systems using MATLAB / Simulink.
- Wired, simulated, and conducted experimental measurements on analog and digital electronic circuits using test benches.

Technologies & Keywords: MATLAB / Simulink, Control Systems, Signal Processing, Instrumentation.

CROSS-FUNCTIONAL SOFT SKILLS

- **Analytical Skills:** Strong theoretical and mathematical foundation from CPGE MP.
- **Teamwork & Rigor:** Proven experience working in scientific pairs and drafting engineering reports.
- **Adaptability:** Continuous technical curiosity and quick adoption of new tools.